

Nabaraj Subedi

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Research Interests: Multimodal Retrieval-Augmented Generation · Agentic AI · 3D World Models · Vision-Language Models · 3D Foundation Models

PUBLICATIONS

ConstructGPT: BIM-Conditioned 3D Diffusion and World Models for Automated Construction Site Monitoring and Compliance Checking 2026

N. Subedi, Y. Majeed, A. Abdelaty, S. V. Sheshappanavar

CVPR 2026 Workshop: 2nd 3D-LLM/VLA Workshop, Accepted, Denver, CO

When More Documents Hurt RAG: Mitigating Vector Search Dilution with Domain-Scoped, Model-Agnostic Retrieval 2026

N. Subedi, A. Abdelaty, S. V. Sheshappanavar Under Review, ACL ARR 2026

Visual-First Multimodal RAG with Agentic Compliance and Visual Grounding: Automating QA and Compliance Checking for Civil Standard Plans 2026

N. Subedi, D. Suvo, A. Abdelaty, S. V. Sheshappanavar (Manuscript under revision in Automation in Construction)

CivilAny: Multimodal Model for Civil Infrastructure Inspection and Design 2026

N. Subedi, A. Abdelaty, S. V. Sheshappanavar (Submitted to ACCV 2026 and under review)

CONFERENCE PRESENTATIONS

ConstructGPT (Poster Presentation) · CVPR 2026 Workshop: 2nd 3D-LLM/VLA Workshop · Denver, CO June 2026

Teaching Models to Work in 3D: Perceiving, Predicting, and Generating 3D Infrastructure Geometry (Lightning Talk) · AI Summit · University of Wyoming Sept 1, 2026

Image Captioning in Nepali Language (Oral) · Gandaki University International Conference (GUIC-

2024) Jan 2024

EDUCATION

University of Wyoming · M.S. Computer Science Expected May 2027

- Advisor: Dr. Shivanand Venkanna Sheshappanavar (Geometric Intelligence Lab, EECS) · Co-advisor: Dr. Ahmed Abdelaty (Civil & Arch. Eng.)
- Courses: Machine Learning & Data Mining, Intro to Large Language Models, Intro to Deep Learning, 3D perception for Robots

Institute of Engineering, Tribhuvan University · B.E. Electronics, Communication & Information Engineering · Full Tuition Scholarship (merit-based) March 2024

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2025 – Present

University of Wyoming · Advisor: Dr. Shivanand Venkanna Sheshappanavar

- Built **ConstructGPT**: BIM-conditioned 3D latent diffusion + construction world model + multimodal RAG compliance agent on H100 GPUs; 0.806 VAE IoU on Rohbau3D, 0.534 leave-one-building-out world-model F_1 , and a 417-sample BIM-anomaly benchmark.
- Developed **MASDR-RAG**: domain-scoped retrieval fixed vector search dilution in 88,907-chunk WYDOT corpus; Precision@10 raised 0.77→0.86 (+11.6%) with 75% latency reduction. 
- Built **Civil-NEXT**, a six-modality any-to-any LLM (text/image/audio/video/3D/CAD, Qwen2.5-72B) with a custom PointCloudOutputProjector; DeepSpeed ZeRO-3 training on 8×H100 over a 737K-sample corpus.
- Built **Visual-First Multimodal RAG** + a 4,056-pair five-DOT benchmark for engineering-plan compliance; benchmarked retrievers from ColPali to 2026 ViDoRe SOTA, achieving **88% zero-shot cross-agency transfer** and **100% verdict accuracy** on 600 CAD compliance drawings via an agentic Planner–Auditor–Synthesizer pipeline. 
- Built **126,463 validated adaptive-step chain-of-thought traces** over **10,172 point clouds** across three 3D benchmarks (scanned grocery items, field crops, synthetic indoor scenes) using Qwen2.5-VL-72B served locally, atop a custom **z-buffered point-cloud renderer** (disk splatting, eye-dome lighting, all-points chunked depth buffer); fine-tuned **PointLLM-7B** and **MiniGPT-3D** on the corpus and benchmarked the against baselines using a locally-hosted Qwen judge.

INDUSTRY EXPERIENCE

Junior Machine Learning Engineer April 2024 – Aug 2024

PalmMind Technology, Nepal

- Built production RAG chatbots for EV dealers and insurance clients using OpenAI, LangChain, and LangGraph with orchestrator-worker workflows; reduced latency and improved multi-hop reasoning.
- Implemented web scraping, OCR (Tesseract), and Redis-based caching for context-aware conversational AI at scale.

Teaching Assistant Sep 2024 – July 2025

Pashchimanchal Campus, Tribhuvan University, Nepal

- Taught C Programming, OOP, Operating Systems, Data Mining undergraduate courses; designed lab activities, guided semester-end projects, and provided feedback to 100+ students.

SELECTED PROJECTS

Modular Real-Time Vision Pipeline for Traffic & Parking Analytics

Summer 2026

Summer Research Project, University of Wyoming

- Built a **pip**-installable vision framework with a six-stage detect→track→render pipeline and four pluggable interfaces (detector/tracker/application/output); parking-occupancy and speed estimation ship as swappable plug-ins.
- Designed speed estimation as two orthogonal axes – distance model (scale/homography) × temporal model (Kalman, windowed median, dual-line trap) – giving **14 estimators from 9 classes**; median smoothing cut speed jitter **32%**.
- Instrumented per-stage latency on a SLURM GPU cluster (L40S): decoupling detection from tracking gave **2× throughput** (54→104 FPS) and **4.7× GPU speedup** over CPU.

MRAG 4.0: Multi-Agent Cognitive Digital Twin for Construction Infrastructure

2025 – Present

Research Project, University of Wyoming

- Integrated site video, CAD drawings, LiDAR terrain data, and operational records into a unified multi-agent reasoning pipeline with specialized agents for visual inspection, CAD intent analysis, and code compliance verification and validated on WYDOT infrastructure data design compliance — capabilities unreachable with standard RAG or existing LLM systems.

WYDOT Multimodal Knowledge Graph Chatbot

Aug 2025 – Present

Research Project, University of Wyoming

- Architected production-grade Multi-Agent RAG system over Neo4j knowledge graph (150K+ nodes, 330K+ relationships) indexing 1,656+ WYDOT engineering documents with hybrid dense+sparse retrieval.
- Deployed serverless backend on GCP Cloud Run + Cloud SQL; built automated feedback-loop ingestion pipeline for self-optimization in production.

Nepali Image Captioning

2023 – 2024

Undergraduate Thesis, Institute of Engineering, Tribhuwan University

- Transformer + Inception V3 for paragraph-length Nepali caption generation on curated 20,350-pair dataset; BLEU-4: 0.59. Published in *Journal of Soft Computing Paradigm* (2024).

TECHNICAL SKILLS

Languages	Python, C/C++, SQL, CUDA
Deep Learning	PyTorch, HuggingFace Transformers, DeepSpeed ZeRO-3, PEFT/LoRA, DiT, Diffusion Models
3D Vision	Open3D, PyTorch3D, Point Clouds
VLM / LLM	Qwen2.5-VL, InternVL, LLaVA, GPT-4V, Fine-tuning, RAG, ColPali, LangChain, LangGraph
Infrastructure	NVIDIA H100/A100, GCP Cloud Run, ARCC HPC Cluster, Docker
Databases	Neo4j (Knowledge Graphs), PostgreSQL, BigQuery

AWARDS & HONORS

Graduate Scholar Award (\$5,000)	School of Computing, University of Wyoming	2026 – 2027
Graduate Research Assistantship (GRA)	University of Wyoming	2025 – Present
Full Tuition Scholarship	Institute of Engineering, Tribhuwan University	2019 – 2024
High School Scholarship	SOS Hermann Gmeiner Secondary School, Gandaki	2017 – 2019